

Water Age Control for Water Distribution Networks via Safe Reinforcement Learning

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Abstract—Reinforcement learning (RL) is a widely used control technique that finds an optimal policy using the feedback of its actions. The search for the optimal policy requires that the system explores a broad region of the state space. This search puts at risk the safe operation, since some of the explored regions might be near the physical system limits. Implementing learning methods in industrial applications is limited because of its uncertain behavior when finding an optimal policy. This work proposes an RL control algorithm with a filter that supervises the safety of the exploration based on a nominal model. The performance of this safety filter is increased by modeling the uncertainty with a Gaussian process (GP) regression. This method is applied to the optimization of the management of a water distribution network (WDN) with an elevated reservoir; the management objectives are to regulate the tank filling while maintaining an adequate water turnover. The proposed method is validated in a laboratory setup that emulates the hydraulic features of a WDN.

Index Terms—Gaussian process (GP), reinforcement learning (RL), safety, water age, water distribution network (WDN), water quality.

I. INTRODUCTION

WATER distribution networks (WDNs) are urban infrastructures that transport drinking water from a water source to numerous end users. The configuration of these infrastructures changes between regions, and the management adapts accordingly to network characteristics, such as the topology of the terrain, the capacity of the water supply, the end-users demand, or the water quality.

Elevated reservoirs are advantageous for certain networks. They can reduce the stress on pumping stations during peak demand by utilizing their storage capacity and ensuring a safe and reliable supply during emergencies or service requirements [1]. Pressure management is a crucial factor in network

maintenance, and elevated reservoirs can help to maintain proper pressure levels within the network. By balancing the pressure, the risk of pipe bursts can be reduced along with ensuring a steady supply of water in case of a power outage [2].

Distribution infrastructures are designed, such that the transportation time from the source to the consumers is low, avoiding water stagnation in the pipes and subsequent deterioration of water quality. Chlorine prevents bacteria growth in the water, and consequently, water quality issues are typically solved by maintaining an adequate chlorine concentration. However, its concentration decays in time [3]. Other types of distribution systems with different water sources can be chlorine-free [4]. Nevertheless, high water age harms water quality in both types, since a long residence time facilitates the growth of bacteria and microfilms. Other factors affect the water quality, such as temperature or mixing in the tank [5]. However, they are out of the scope of this work, as their effect is negligible. This work studies the operation of small water utilities in Denmark, where the average temperature is below 20 °C, and the mixing inside the tank is sufficient, so its impact on the deterioration of the water quality is considered negligible [6]. Therefore, the main factor affecting the water quality in the studied network is the residence time in the reservoirs.

A. Control of WDNs

Adequate management of a WDN comprises four main objectives that [7] defines as smoothness of the control actions, the safety of the supply, economy, and water quality.

A WDN with elevated reservoirs is often managed using rule-based policies. These policies rely on minimum and maximum tank levels for actuating the pumps, which regulate the tank filling. The water age in the tank is typically not monitored in real time, and the issues related to water age are solved by manually scheduling an emptying of the storage elements. Although rule-based approaches offer the benefits of robust and easy-to-maintain management, they can overlook important objectives, such as smoothness and economics. Moreover, relying on a manual process to renovate stored water raises the risk of lower water quality and decreased efficiency.

The efficient management of a WDN is a well-established research area that includes many different approaches, most of

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them in the framework of optimal control [8], [9]. However, the good performance of these techniques relies on a system model that requires continuous calibrations. The maintenance of system models and controllers requires qualified personnel that is not always affordable, especially for small water utilities. The model dependence and high commissioning costs motivate the use of data-driven techniques that replace the reliance on detailed system knowledge and facilitate the implementation of optimal control techniques in a broader range of utilities. Some research utilizes MPC methods, which are coupled with other control techniques, such as chance constraints [10], robust control [11], or stochastic methods [12], to tackle issues related to model dependency.

B. Motivation and Contribution of the Research

Reinforcement learning (RL) is a control approach that relies on measured data instead of a specific model to find an optimal policy, and RL gradually learns from the feedback of its control actions. Applications of RL algorithms have been successful in controlling dynamical systems [13], [14], [15]. However, its use in industrial applications is still limited. The process of learning an optimal control policy includes exploring a wide range of state space, which can result in lengthy training times or unsafe operating conditions. Therefore, learning while having an optimal operation is conflicting objectives that must be balanced.

The algorithm presented in [16] shows a data-driven method that similarly combines the open-loop and close-loop trajectories with linear model assumptions for the reduction of the training time. Several studies have explored different approaches to enhance the safety of learning controllers. Some approaches incorporate safety criteria into the optimization process, using methods, such as barrier functions [17]. Other studies have focused on ensuring a safe exploration process by utilizing regressions that support real-time control. For instance, some studies use a nominal linear system for trajectory planning [18], [19], and others combine the regression of a nominal model with a Gaussian process (GP) model to account for uncertainty [20], [21]. The approach presented in [22] introduces a combination of cooperative game theory, GP, and differential dynamic programming to provide robust optimization in systems with Poisson and Gaussian disturbances. Another safety approach combines a safe optimization framework from MPC with the learning capabilities of RL in [23].

This article presents an optimal control solution for a WDN with the elevated reservoirs that require minimal system knowledge. Given that a WDN is a critical infrastructure that must operate continuously, the focus of this approach is on ensuring a safe operation of the system over achieving optimal performance. The proposed control strategy regulates the tank filling and maintains the water age below safety limits. The control architecture consists of a combination of RL control and a safety filter. The safety filter supervises the learned policy and maintains local robustness near the safety boundaries. The proposed safety filter is formed by a combination of a deterministic nominal model that includes

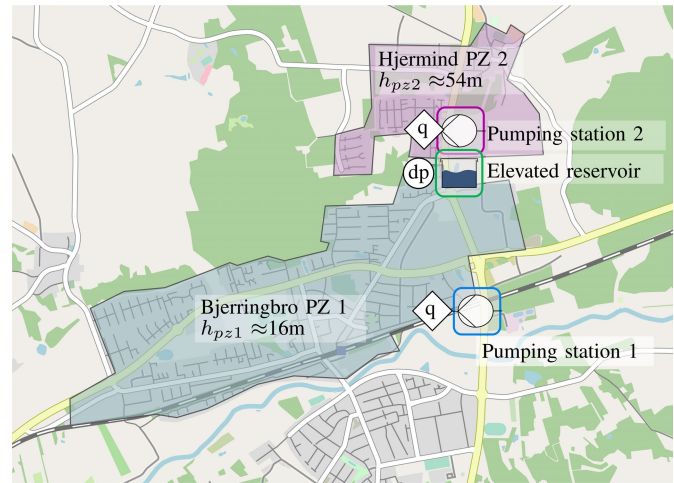


Fig. 1. Map of a WDN with an elevated reservoir in Denmark. PZ1 covers the Bjerringbro district, and PZ2 covers Hjerlind, which is located at a different elevation (h_{pz}).

inaccurate system knowledge and GP regression, which models the system uncertainties.

C. Organization of This Article

The remainder of this article is organized as follows. Section II describes the system model of a WDN with an elevated reservoir. The model includes the tank dynamics and the turnover of the tank. Section III develops the model-free control strategy. Section IV develops the safety filter that supervises the learning policy. Section V presents the case study and the validation with experimental results. Section VII summarizes the contribution of the work and introduces some ideas for future development of the proposed method.

II. SYSTEM MODEL

This section describes a model of a small WDN with an elevated reservoir. This work uses the network configuration of Bjerringbro to validate the usability of the control scheme. This distribution topology is typically found in small water utilities; see Fig. 1. This article considers a noninterrupted (continuous) operation where the pipe network is designed to avoid water stagnation. However, a significant water quality risk comes from the water age in the storage tanks. The turnover in the tank is used to monitor the water age. Therefore, a model of the daily turnover of the reservoir is developed based on available network measurements. The topology of this system consists of two pressure zones (PZs), and an elevated reservoir, a simplified map, and configuration scheme of the network are illustrated in Figs. 1 and 2, respectively. The pumping station regulates the inflow q_{p1} , this signal is the control variable for the controller design in this work, and the inflow q_{p2} is determined by the demands at PZ2 and the pressure at the boosting station; therefore, this signal is not a control variable considered in the controller design.

A. Water Distribution Network

WDNs with elevated reservoirs are stiff systems where the network's flow dynamics are much faster than the elevated

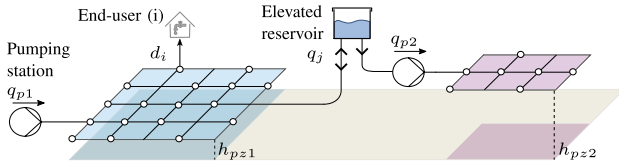


Fig. 2. Scheme of the study case network topology, a WDN with a single pumping station and an elevated reservoir. It is divided into PZ1 and PZ2 (blue and pink), and PZ2 is supplied through a booster station that regulates the pressure. The nodal demand d_i represents the consumption from the end user i , and q_j represents the inflow to the tank in connection j .

reservoir dynamics. The pump and end-user flows are assumed to be ideally regulated, and the system model is simplified by focusing on the dynamics of the tanks, as described in [24]

$$A_{er}\dot{h} = \sum_{j=1}^{n_{er}} q_j(t) \quad (1)$$

where A_{er} is the cross-sectional area of the elevated reservoir, n_{er} is the number of inlets in the reservoir, and q_j is the flow at the reservoir inlet j . Due to mass conservation in the network, the relation between the supply flows q_{p1} and q_{p2} , the demand flows d_i , and reservoir flows is described by

$$\sum_{j=1}^{n_{er}} q_j(t) = q_{p1}(t) - \sum_{i=1}^{n_d} d_i(t) - q_{p2}(t) \quad (2)$$

where $d_i > 0$ represents the demand of end user in the PZ1, n_d is the number of end users, and q_{p2} is the flow to PZ2. The demand pattern between the different end users is similar; therefore, the total demand $\bar{d}$ of a PZ is described by

$$d_i(t) = \beta_i \bar{d}(t) \quad \forall i = 1, \dots, n_d \quad (3)$$

where β_i is a scaling factor. In a real WDN, the total water demand can be described as a stochastic process that follows a daily pattern. In this work, the signal described by the mean value variation of the stochastic process is approximated with a Fourier series (FS) of the N th order

$$\bar{d}(t) = a_0 + \sum_{n=1}^N (a_n \cos(\omega_n t) + b_n \sin(\omega_n t)) + w \quad (4)$$

where a_0, a_n , and $b_n \in \mathbb{R}$ are the Fourier coefficients, $\omega_n = 2\pi n f_0$ and f_0 represents the fundamental frequency, and w is Brownian noise.

Finally, the continuous-time models of the tank (1) and total demand (4) are transformed into discrete-time state-space models using zero-order hold; first, the tank model is defined as follows:

$$h_{k+1} = f(h_k, u_k, \bar{d}_k) = Ah_k + Bu_k + E\bar{d}_k \quad (5)$$

where $h_k \in \mathbb{R}$ is the system state; $u_k \in \mathbb{R}$ is the controlled input flow; $\bar{d}_k \in \mathbb{R}$ represents the total system disturbances; and A, B , and E are constant matrices. Then, the periodic disturbance model is also expressed in a state-space discrete form

$$\begin{aligned} s_{k+1} &= A_d s_k \\ \bar{d}_k &= C_d s_k \end{aligned} \quad (6)$$

where the system matrix $A_d = \text{diag}(1, F_1, \dots, F_N)$, with

$$F_n = \begin{bmatrix} \cos(\omega_n \Delta t) & -\sin(\omega_n \Delta t) \\ \sin(\omega_n \Delta t) & \cos(\omega_n \Delta t) \end{bmatrix}$$

where Δt is the sampling time and the output matrix C_d includes the Fourier coefficients. The state vector $s_k \in \mathbb{R}^{n_s}$, with $n_s = 2N + 1$, is subjected to the following initial condition:

$$s_{i,t_0} = \begin{cases} c_0, & \text{if } i = 0 \\ \cos(\omega_n t_0), & \text{if } i > 0, \quad i \text{ odd} \\ \sin(\omega_n t_0), & \text{if } i > 0, \quad i \text{ even} \end{cases} \quad (7)$$

where c_0 is a constant, t_0 is the initial time value, and the index vector $i \in \mathbb{Z}, [0, n_s]$.

B. Estimating the Water Age From Measurements

The real-time monitoring of water quality and its regulation is complex, since it is affected by multiple factors, such as biological and chemical composition, mixing, and temperature [1]. This work aims to control the quality of the stored water by only having adequate storage management. Therefore, it focuses only on those factors that the network management can control. Assuming a low water temperature and ideal mixing in the reservoir, the water quality is homogeneous for the stored volume. Thus, the water age, or average residence time (ART), becomes the principal factor that impacts water quality. The ART in an elevated reservoir is defined as follows [25]:

$$\text{ART}_k = \frac{v_{av,k}}{q_{av,k}}, \quad \text{for } q_{av,k} > 0 \quad (8)$$

where $v_{av,k}$ is the average volume and q_{av} is the average flow entering the reservoir. The study presented in [26] suggests a 3–5-days complete water turnover as a basis; nevertheless, each utility must adapt the management to meet its own quality requirements. The flow measurements in a WDN are limited, and the proposed network only has flow measurements at the pumping stations. The flow measurements at the reservoir are not available, but they can be inferred from the reservoir level due to mass conservation. Therefore, the ART is reformulated to be computed from level measurement as follows. First, by discretizing the reservoir dynamics (1), the average volume is defined with previous level measurements

$$v_{av,k} = \frac{A_{er} \sum_{i=k-n_{av}}^k h_i}{n_{av}} \quad (9)$$

where n_{av} represents the number of samples per period. Second, the average inflow is denoted as follows:

$$q_{av,k} = \frac{A_{er} \sum_{i=k-n_{av}}^k \delta_i}{n_{av}} \quad (10)$$

where δ represents only the positive level variations

$$\delta_{i+1} = \max\{(h_{i+1} - h_i), 0\} \quad (11a)$$

$$= \max\left\{\left(A_{er}^{-1} \sum_{j=1}^{n_{er}} q_{j,i} \Delta t\right), 0\right\} \quad (11b)$$

where $q_{j,i}$ is the flow at inlet j at time i . Note that (11) has two expressions depending on the available measurement; for notation simplicity, it is represented as a function of the reservoir flows $\delta(q_k)$, as shown in (11b). Finally, the daily volume turnover [%] is denoted as a function of the flows in the network and the level of the reservoir

$$\tau_k = g(h_k, q_k) = 100 n_{av} \frac{q_{av,k}}{v_{av,k}}. \quad (12)$$

For numerical convenience, this article uses the reservoir turnover $\hat{\tau}$ over ART to regulate the aging of the water in the reservoir.

1) *Incremental Average Approximation:* The turnover defined in (12) is a nonlinear equation that depends on past measurements (time lag), and this requires large data storage. This article proposes an approximation by using an incremental average to reduce the data storage and computational load. First, denote the mean of the level variations δ

$$m_k(q_k) = \frac{1}{n_{av}} \sum_{i=k-n_{av}}^k \delta_i(q_k). \quad (13)$$

Then, the new mean is computed by extending (13) and including the new input (11) as follows:

$$\hat{m}_{k+1} = \frac{1}{n_m} (\delta_{k+1}(q_k) + (n_m - 1)\hat{m}_k) \quad (14a)$$

$$= \hat{m}_k + \frac{\delta_{k+1}(q_k) - \hat{m}_k}{n_m} \quad (14b)$$

where n_m is the window length of the moving average filter. Subsequently, the daily turnover output is approximated as follows:

$$\hat{\tau}_k = \hat{g}(q_k) = 100 \frac{\hat{m}_k}{h^*} \quad (15)$$

where h^* is a constant representing the average level during steady-state operation and q_k is the sum of reservoir inflows. The computational implementation of this model uses expression (14b).

2) *Topology Considerations:* This section transforms the computation of the water turnover in the tank for different hydraulic configurations. The model presented in (11) computes positive level variations in a reservoir. However, this model is only accurate for tanks with a single inlet.

This issue motivates an alternative formulation of the turnover model that describes the infeed of freshwater for multiple inlets.

Consider an example with the hydraulic configuration illustrated in Fig. 2, where the reservoir has two pipe connections, one bidirectional and one outflow. This configuration ensures a regular inflow of freshwater during steady-state conditions via inlet 1. However, the model (11), that utilizes level measurements or the sum of the flows, returns zero level variations, since $\sum_{j=1}^{n_{er}} q_j(t) = 0$.

This model aims to calculate the volume of freshwater introduced to the tank in a given time. For this, expression (1) is expanded for different reservoir inlets and integrated over a

time period T as follows:

$$A_{er} \int_T \dot{h} dt = \int_T \underbrace{A_{er} \dot{h}_1}_{\text{inlet1}} dt - \int_T \underbrace{A_{er} \dot{h}_2}_{\text{inlet2}} dt \quad (16)$$

where $\dot{h}_1$ and $\dot{h}_2$ represent the level rate from inlet 1 and inlet 2, respectively. By analyzing this particular study case, only inlet 1 inflows water to the reservoir, and inlet 2 is a fixed outfeed but its magnitude is known (measured). The level rate $\dot{h}_1$ cannot be measured, but it can be computed with the available measurements. With this system knowledge, expression (16) is discretized and represented in terms of reservoir level and PZ2 flow

$$\Delta h_1 = \Delta h + A_{er}^{-1} q_{p2} \Delta t \quad (17)$$

where $\Delta h = h_{k+1} - h_k$. Then, the positive level variations are computed with $\dot{h}_1$

$$\delta_{k+1} = \max\{\Delta h_1, 0\}. \quad (18)$$

Subsequently, the turnover (15) is similarly computed with inlet 1 positive level variations (18).

III. RL CONTROL

This section presents the formulation of an optimal-adaptive controller, which adapts to the hydraulic configuration and the consumption pattern of the network to achieve the utility management objectives. In the proposed control structure, the RL control aims to regulate the tank filling and the smoothness of the control. Water quality is later formulated in Section IV as a safety problem.

A. Augmented State Space

This work uses the augmented state-space model proposed in [24] to construct the learning scheme. This model combines the discrete-time tank model (5) and the periodic disturbances (6). The states s are included in the model to enable identification of the periodic disturbances

$$\begin{bmatrix} h_{k+1} \\ s_{k+1} \end{bmatrix} = \begin{bmatrix} A & EC_d \\ \mathbf{0} & A_d \end{bmatrix} \begin{bmatrix} h_k \\ s_k \end{bmatrix} + \begin{bmatrix} B \\ \mathbf{0} \end{bmatrix} u_k \quad (19)$$

where u_k is the controlled input; in the study case, q_{p1} is the only controlled input. This work assumes that all the states are measurable, h is measured with a sensor, and s is generated using known periodicity. Finally, by defining the state vector $x_k = [h_k \ s_k^T]^T$, system (19) is represented in a compact form

$$x_{k+1} = A_e x_k + B_e u_k \quad (20)$$

where $x \in \mathbb{R}^{m_a}$ and $u \in \mathbb{R}^{n_a}$. The feedback control policy is given by the following linear controller:

$$u_k = \pi(x_k) = -K x_k. \quad (21)$$

B. Cost Function and Bellman Equation

A value function that includes the system objectives with a discount factor is defined as follows:

$$V(x_k) = \sum_{i=k}^{\infty} \gamma^{i-k} \rho(x_i, u_i) \quad (22)$$

where $0 < \gamma < 1$ is a constant factor that discounts the rewards obtained in the future. The instant reward ρ_k includes the application objectives, and it is denoted as follows:

$$\rho(x_k, u_k) = (x_k - x^*)^T Q_e (x_k - x^*) + u_k^T R u_k \quad (23)$$

where $Q_e > 0$ and $R > 0$ are constants that penalize the tracking error and high control actions, respectively, and $x^* \in \mathbb{R}^{m_a}$ includes the reference of the tank h^* .

By formulating the previous cost function with Bellman's optimality principle, the optimal value function is expressed with the Bellman equation as presented in [27]

$$V^*(x_k) = \min_u (\rho(x_k, u_k) + \gamma V^*(x_{k+1})) \quad (24)$$

with the notation $(\cdot)^*$ representing the optimal value. Following the approach described in [24], a candidate parametrization of the value function (24) is

$$V(x_k) = x_k^T P x_k + G x_k + c. \quad (25)$$

Then, combining (24) and (25) leads to

$$V^*(x_k) = \min_u (\rho(x_k, u_k) + \gamma (x_{k+1}^T P x_{k+1} + G x_{k+1} + c)). \quad (26)$$

Finally, the previous V-value function (26) is expressed as a Q-value function, and it is represented in terms of state x and control action u , subsequently where the system dynamics of the augmented state-space model (19) are introduced

$$\begin{aligned} Q(x_k, u_k) &= (x_k - x^*)^T Q_e (x_k - x^*) + u_k^T R u_k \\ &\quad + \gamma [(Ax_k + Bu_k)^T P (Ax_k + Bu_k) \\ &\quad + G(Ax_k + Bu_k) + c]. \end{aligned} \quad (27)$$

Then, by rearranging (27) into a matrix form

$$\begin{aligned} Q(x_k, u_k) &= \begin{bmatrix} x_k \\ u_k \end{bmatrix}^T \begin{bmatrix} M_{xx} & M_{xu} \\ M_{ux} & M_{uu} \end{bmatrix} \begin{bmatrix} x_k \\ u_k \end{bmatrix} \\ &\quad + \begin{bmatrix} x_k \\ u_k \end{bmatrix}^T \begin{bmatrix} N_x \\ N_u \end{bmatrix} + \begin{bmatrix} N_x \\ N_u \end{bmatrix}^T \begin{bmatrix} x_k \\ u_k \end{bmatrix} + d. \end{aligned} \quad (28)$$

Then, the optimal control policy for (28) is calculated as follows:

$$u_k^* \in \arg \min_u Q(x_k, u_k) = M_{uu}^{-1} (M_{ux} x_k + N_u). \quad (29)$$

C. Q Value Using Linear Architecture

Section III-B presents a Q-value function for a linear system that is built with the system matrices A_e and B_e . Section III-C presents an equivalent control approach where the Q-value function is approximated, since the system matrices are unknown. This model-free approach proposes a linear parametric approximation for the Q-value function of the form

$$\hat{Q}(x_k, u_k) = \phi^T(x_k, u_k) \theta \quad (30)$$

where $\phi \in \mathbb{R}^{n_b}$ is a column vector with the basis functions (BFs) and $\theta \in \mathbb{R}^{n_b}$ is the coordinate vector with the number of bases $n_b = (m_a + n_a + 1)(m_a + n_a)/2$. The column vector is built with a finite set of polynomials of the 2nd degree, and this polynomial approximation is inspired by the quadratic form of its model-based version (28)

$$\phi(x_k, u_k) = [x_{1,k}^2, x_{1,k} x_{2,k}, \dots, x_{m_a,k}^2, x_{m_a,k} u_k, u_k^2]^T. \quad (31)$$

Subsequently, the optimal control law for the approximated Q-value function is determined by

$$u_k \in \arg \min_u \hat{Q}(x_k, u_k) = \arg \min_u \phi^T(x_k, u_k) \theta. \quad (32)$$

According to [27], the optima control input found by solving (32) is the linear controller

$$u_k = \hat{\pi}(\theta, x_k) = \hat{K}(\theta) x_k. \quad (33)$$

D. Parameter Update

The vector θ contains the parameters that characterize the Q-value function. This vector is initially unknown, and its parameters have to be identified iteratively by using collected data. A temporal difference (TD) algorithm is used for approximating the Q-value function, and this algorithm minimizes the TD error between successive iterations [28], [29]. The TD with function approximators is formulated as follows:

$$\begin{aligned} \phi^T(x_k, u_k) \theta_{k+1} &= (1 - \alpha) \phi^T(x_k, u_k) \theta_k \\ &\quad + \alpha [\rho(x_k, u_k) + \gamma \phi^T(x_{k+1}, u'_k) \theta_k] \end{aligned} \quad (34)$$

where $0 < \alpha < 1$ is a constant learning rate. This method uses batch learning to minimize the TD, and it consists of applying a control policy and collecting the measured data. When a batch of data is completed with m iterations, (34) is solved using least squares as follows:

$$\theta_{l+1} = (1 - \alpha) \theta_l + \alpha (\Phi_l \Phi_l^T)^{-1} \Phi_l [J_l + \gamma \Phi_l^T \theta_l] \quad (35)$$

where l is the iteration number, and $\Phi_l = [\phi_1, \dots, \phi_{l+m}]$ and $J_l = [\rho_1, \dots, \rho_{l+m}]^T$ are a matrix and a vector generated by evaluating the collected data into the polynomial BFs (31) and reward functions (23), respectively.

IV. POLICY SUPERVISOR

The controller designed in Section III provides an optimal control policy for a continuous state-action space. This policy is obtained from a model-free optimization where no constraints are considered. However, the operation of physical systems must be restricted to certain areas that guarantee a safe operation. Therefore, a policy supervisor module, also referred to as a safety filter, is introduced in the control structure [19]; see Fig. 3. This module aims to assess the safety of the learned policy, first by predicting a potential violation of the safety regions and then to provide a safe action. Moreover, this filter is used to restrict the water age in the tank, and it acts as a fallback control when the safety limits are violated.

This article presents two approaches to evaluate the system's safety: one is based on a nominal model, and the other is based on a combined model that consists of a nominal model and a GP regression.

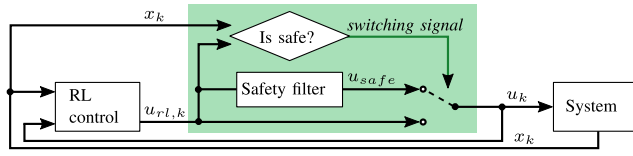


Fig. 3. Block diagram of the control architecture where the RL control is connected in series with a policy supervisor (green). The policy supervisor switches the control action based on a 1-step ahead prediction.

A. Safety Regions

In this work, the WDN's safety is built around two factors, the physical limitations of the tank and the water aging in the tank. The goal of this filter is to repel the operation from these unsafe areas while taking into consideration the control policy (33) that now is denoted by u_{rl} . In this article, the safety regions are defined as follows:

$$h = \mathcal{H} \triangleq \{h \in \mathbb{R}^{n_h} | \underline{h} \leq h \leq \bar{h}\} \quad (36a)$$

$$\tau = \mathcal{T} \triangleq \{\tau \in \mathbb{R}^{n_\tau} | \underline{\tau} \leq \tau \leq \bar{\tau}\} \quad (36b)$$

$$u = \mathcal{U} \triangleq \{u \in \mathbb{R}^{n_u} | \underline{u} \leq u \leq \bar{u}\} \quad (36c)$$

where the notations $(\cdot)$ and $(\bar{\cdot})$ define lower and upper bounds, respectively.

B. Nominal Filter

The filter approach uses a nominal model to predict potential risks in operation and provide a safe action if necessary. The nominal model is a linear model that is built with a priori system knowledge, and therefore, it does not describe the exact behavior of the system. However, it represents the essential system dynamics necessary to correct the path.

1) *Nominal Model*: This article refers as a nominal model, an imperfect version of the model (5), and a linear model of the form

$$\hat{h}_{k+1} = \hat{f}(h_k, u_k) = \hat{A}h_k + \hat{B}u_k + \hat{E}d_{av} \quad (37)$$

where d_{av} is a constant representing the average of the total demand, and $\hat{A}$, $\hat{B}$, and $\hat{E}$ are system matrices of the nominal model. The turnover signal is represented by the algebraic equation (15). This signal is computed with the flow model. Although the tank inflow is not directly measured, q_k is inferred from the network flows: pump inflow 1, u_k (decision variable); the average demand, d_{av} (nominal guess); and pump inflow 2, q_{p2} (measured).

2) *Nominal Safe Policy*: The safety filter is formulated as a constraint optimization problem

$$u_{safe} \in \arg \min_{u_k} \|u_{rl,k} - u_k\|_{Q_1}^2 + \|\tau^* - \hat{\tau}_k\|_{Q_2}^2 + \|\xi\|_S^2 \quad (38a)$$

$$\text{s.t. } \hat{h}_{k+1} = \hat{f}(h_k, u_k) \quad (38b)$$

$$\hat{\tau}_k = \hat{g}(q_k) \quad (38c)$$

$$\hat{h}_k \geq \underline{h} - \xi_1 \quad (38d)$$

$$\hat{h}_k \leq \bar{h} + \xi_1 \quad (38e)$$

$$\hat{\tau}_k \geq \underline{\tau} - \xi_2 \quad (38f)$$

$$\underline{u} \leq u_k \leq \bar{u} \quad (38g)$$

$$\xi \geq 0. \quad (38h)$$

Algorithm 1 Safe LS-TD for Q Function—Nominal Filter

```

1: Input:  $\gamma, \alpha, n_s$ ,
2: Initialisation:  $l \leftarrow 0, x_0, \theta_0$  where  $\hat{\pi}(\theta_0)$  must be an
   admissible policy.
3: repeat at every iteration  $k = 0, 1, 2, \dots$ 
4:   apply  $u_k$  and measure  $x_{k+1}$ 
5:    $\Upsilon_{l,s} \leftarrow \rho(x_k, u_k) + \gamma \hat{Q}(x_{k+1}, \hat{K}_l x_{k+1})$ 
6:   if  $k = (l+1)n_s$  then ▷ Policy update
7:      $\theta_{l+1} \leftarrow (1 - \alpha)\theta_l + \alpha(\Phi_l \Phi_l^T)^{-1} \Phi_l \Upsilon_l$ 
8:      $\hat{\pi}(\theta_{l+1}, x) \leftarrow \arg \min_u \phi(x, u)^T \theta_{l+1}$ 
9:      $l \leftarrow l + 1$ 
10:  end if
11:  if  $\hat{h}_{k+1} \in H$  and  $\hat{\tau}_k \in \mathcal{T}$  then ▷ Policy supervisor
12:     $u_k = \hat{K}(\theta_l)x_k + \epsilon_k$ 
13:  else
14:     $u_k = u_{safe} + \epsilon'_k$ 
15:  end if
16: until

```

The objective function (38a) consists of two terms, the first penalizes the difference between the safe and the learning action, u_{rl} , and the second term represents the tracking error between a constant reference value, τ^* , and the turnover of the tank. Q_1 and Q_2 are constants penalizing the aforementioned terms. ξ is a slack variable that relaxes some of the inequality constraints.

Algorithm 1 is executed in real time, and it comprises two main objectives, updating the RL policy and rectifying the policy during potential unsafe operation [19]. Remark that the control action in the algorithm consists of the policy and a persistence of excitation signal ϵ_k . This signal facilitates that the collected data contain sufficient information to identify the Q-value function. Safety and learning are conflicting, since a high gain in the persistence excitation signal might break the safety guarantees, or, on the other hand, strict safety policies might lower the quality of the collected data. Note that a different persistence of excitation signal ϵ'_k is added to the safety actuation. The magnitude of this signal is adjusted to have a low impact on safety but is sufficient to avoid numerical issues during the parameter identification. This work tolerates the risk of unsafe operation around the boundaries for the benefit of the overall performance. In this way, the disruptive effect of the safety actions in the learning is reduced.

C. Combined Filter

The knowledge of the system is limited in industrial applications, and the filter can become either conservative or reckless depending on how accurate is the system information. The combined filter aims to improve the performance of the previous safety method by reducing the reliance on the calibration of the nominal model. This method for compensating the model uncertainty consists of an imperfect linear model (nominal) and a GP regression. The GP regression term is added to the system dynamics to capture the deviation of the nominal model from the real system.

1) *Combined Model*: Consider that the discrete system model is formed by two terms as follows:

$$h_{k+1} = \hat{f}(h_k, u_k) + B_r(r(z_k) + w_k) \quad (39)$$

where $\hat{f}$ represents the known nominal model and r the nonmodeled dynamics of the system; both functions $\hat{f}$ and r are assumed to be differentiable. The observed input vector is built with the augmented system states and control action, $z_k = [x_k^T, u_k^T]^T$, B_r is an index matrix and the random variable, and $w_k \sim \mathcal{N}(0, \Sigma^w)$ is i.i.d. process noise.

The residual function $r(z)$ is unknown; however, by using a GP regression, $r(z)$ can be inferred. Then, by combining the nominal model with the GP approximation $\hat{r}(z)$, the system model (39) is transformed into

$$h_{k+1} \approx \hat{f}(h_k, u_k) + B_r \hat{r}(z_k). \quad (40)$$

The derivation of the GP approximation $\hat{r}(z)$ with predictive mean and variance equations is given in the Appendix; then, the implementation of the GP training is briefly described.

Subsequently, to compensate the uncertainty in the turnover, this approach considers the level model $\tilde{h}$ to compute (42c).

2) *Combined Safe Policy*: To introduce the residual function (40) into the safety filter, a reformulation of the safe-optimal control problem is required. In this way, the constraints are represented as chance constraints as follows:

$$\Pr\{h \in \mathcal{H}\} \geq p_h \quad \forall k \quad (41a)$$

$$\Pr\{\tau \in \mathcal{T}\} \geq p_\tau \quad \forall k \quad (41b)$$

where p_h and p_τ are the satisfaction probabilities. Finding an algebraic solution to the problem is difficult when working with chance constraints. Therefore, a transformation of the chance constraint (41) into deterministic equivalents (42d)–(42f) is performed. Then, by utilizing the mean and variance models from (50), the optimization problem is formulated as follows:

$$u_{\text{safe}} \in \arg \min_{u_k} \|u_{r,l,k} - u_k\|_{Q_1}^2 + \|\tau^* - \hat{\tau}_k\|_{Q_2}^2 + \|\xi\|_S^2 \quad (42a)$$

$$\text{s.t. } \tilde{h}_{k+1} = \hat{f}(h_k, u_k) + B_r \mu_k^r(z_k) \quad (42b)$$

$$\hat{\tau}_k = \hat{g}(\tilde{h}) \quad (42c)$$

$$\tilde{h}_{k+1} \geq \underline{h} + K_c \sigma^r(z_k) - \xi_1 \quad (42d)$$

$$\tilde{h}_{k+1} \leq \bar{h} - K_c \sigma^r(z_k) + \xi_1 \quad (42e)$$

$$\hat{\tau}(\tilde{h}_{k+1}) \leq \underline{\tau} - \xi_2 \quad (42f)$$

$$\underline{u} \leq u_k \leq \bar{u} \quad (42g)$$

$$\xi \geq 0 \quad (42h)$$

where the standard deviation $\sigma^r(z_k) = (\Sigma^r)^{1/2}(z_k)$ is computed with the variance model (50c), K_c represents the confidence gain, and ξ is a slack variable that relaxes some of the inequality constraints. Remark that the mean $\mu_k^r(z_k)$ and variance function $\Sigma_k^r(z_k)$ include the decision variable u_k in its input vector z_k .

Algorithm 1 is executed in real time with the combined safety filter. Then, a prediction is considered safe if meets the

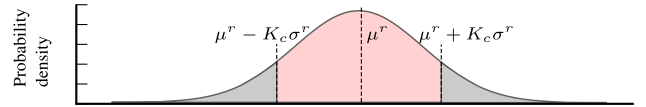


Fig. 4. Scheme of a normal distribution where red represents the area inside the CI of the residual function.

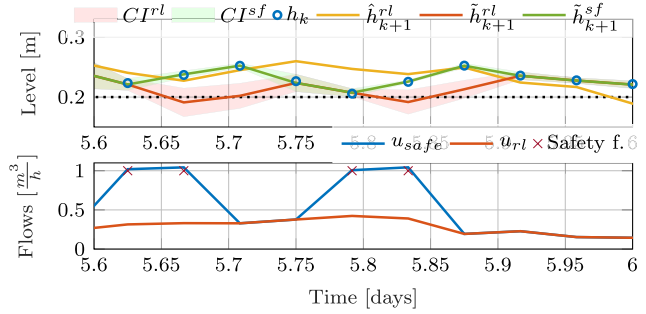


Fig. 5. Simulation of the WDN with an RL controller and a combined filter. Top: blue dots represent the real level measurement, and the dotted line represents the lower boundary. The nominal model signal $\hat{h}_{k+1}^{rl} = \hat{f}(h_k, u_{rl})$ and the combined model $\tilde{h}_{k+1}^{rl} = \hat{f}(h_k, u_{rl}) + B_r \hat{r}(z'_k)$ are 1-step ahead predictions, and $\tilde{h}_{k+1}^{sf} = \hat{f}(h_k, u_k) + B_r \hat{r}(z_k)$ is the safe prediction. Bottom: RL control input u_{rl} and the control input for safe exploration u_{safe} .

following criteria:

$$\underline{h} + K_c \sigma^r(z'_k) \leq \tilde{h}_{k+1}(z'_k) \leq \bar{h} - K_c \sigma^r(z'_k) \quad (43a)$$

$$\underline{\tau} \leq \hat{\tau}(\tilde{h}_{k+1}) \quad (43b)$$

where the observed input vector z'_k is built with the augmented system states and the RL control action, $z'_k = [x_k^T, u_{rl,k}^T]^T$. Fig. 4 illustrates a normal distribution and how the deviation from the mean value is adjusted with K_c . Remark that this method must fulfill

$$|K_c \sigma^r(z_k)| \leq |\bar{h} - \underline{h}|/2. \quad (44)$$

This condition limits the width of the confidence interval (CI) based on the distance between upper and lower bounds. The standard deviation, σ^r , is time-variant, and its regression model is trained online. This challenges the selection of a suitable gain. Therefore, this article proposes the use of a naive prediction [30] as a protection mechanism that makes the safety problem less sensitive to the GP model. It consists on canceling the variance term in (42d) and (42e) by setting $K_c = 0$ when condition (44) is not satisfied. Fig. 5 shows a period of a simulation where the safety filter is intermittently rectifying the system trajectory. The top graph shows the free-exploration prediction $\tilde{h}_{k+1}^{rl}$ with CI^{rl} ; they are computed with the measured state h_k and learning control input u_{rl} , and this signal is used for detecting unsafe trajectories. The safe-exploration prediction is represented by $\tilde{h}_{k+1}^{sf}$, and it is computed with the state h_k and the safe control input u_{safe} ; when no risk of constraint violation is predicted, both signals are aligned. The bottom graph shows the control signals and how the safety filter corrects the control action u_{rl} when the prediction crosses the boundaries. Furthermore, a clear deviation of $\tilde{h}_{k+1}^{rl}$ is observed with respect to $\tilde{h}_{k+1}^{sf}$, and this is due to the poor calibration of the nominal model. The GP model of the uncertainty is trained online, and the CIs are gradually reduced.

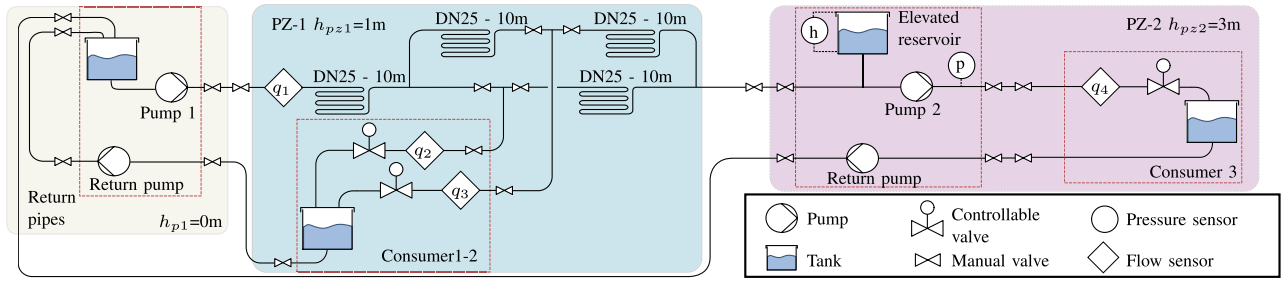


Fig. 6. Scheme of the laboratory setup with a main pumping station (gray), two PZs (PZ1—blue and PZ2—pink), and an elevated reservoir with a booster station (pump 2). The water is collected at the consumer tanks and recirculated to the supply reservoir. A legend is shown on the bottom-right side.

V. RESULTS

The proposed control architecture is validated in a laboratory testbed, first with the nominal and then with the combined safety. The emulated network represents a WDN with an elevated reservoir and two PZs, reproducing the network features of the distribution system shown in Fig. 1.

A. Study Case: Bjerringbro

A modular laboratory testbed is built at the Smart Water Infrastructures Laboratory (SWIL), Aalborg University, where the hydraulic configuration of the study case is emulated in a smaller scale.

The actual network and testbed are divided into two PZs; PZ-1 is supplied directly from pumping station 1 and elevated reservoir, and PZ-2 is located at a higher elevation and is supplied from the elevated reservoir with pumping station 2; see Fig. 6. In this testbed, the elevation at the different network areas is emulated by locally controlling the air pressure at the consumption nodes (collection tanks). Note that the hydraulic configuration of the laboratory elevated reservoir, with one inlet, differs from reality, which has two separated inlets (see Fig. 2). This connection makes a major difference in a real network. However, for this experiment, the impact of this connection variation is negligible, since the quality of the water is not monitored, and the hydraulic properties are equivalent.

The consumption profiles are emulated with an FS of second order, which approximates the consumption pattern from the entire PZs. The magnitude of the signals is reduced to fit the testbed scale, and the consumption ratio between PZ1 and PZ2 is kept between 0.73 and 0.27, respectively, reproducing the consumption between areas. The time scale of the tests is adapted to perform accelerated test, from hours to minutes. Pump 2 has a local pressure control that boosts the pressure at the PZ-2 consumer. Finally, the water is transported to the supply reservoir for recirculation.

In addition, local PI controllers regulate the pump inflow at the supply nodes and the valve outflow at the demand nodes. To reduce the impact of these local controllers in the tests, the sampling time for local controllers is 1 s, and for supervisory control (RL) is 60 s.

The testbed is equipped with multiple sensors. However, only three measurements are available for the supervisory control, the flow at the pumping units (q_{p1} and q_{p2}), and the differential pressure (h) at the elevated reservoir. The rest of the sensors are used for hardware protection and for

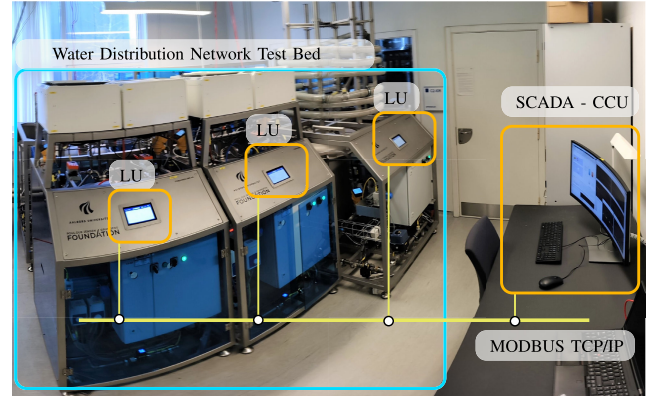


Fig. 7. Picture of the testbed at the SWIL. Left: laboratory modules emulating a WDN are placed. Right: SCADA PC for monitoring and control of the testbed. The communication architecture between CCU and LUs is represented in yellow.

monitoring the testbed. The data from the testbed are locally collected at the local units (LUs) with Codesys Runtime [31], and it is interfaced with the central control unit (CCU) via TCP/IP Modbus. The proposed control strategy is implemented in Simulink at the CCU. A simple representation of the communication architecture is shown in Fig. 7, and a detailed explanation of the laboratory system is presented in [32].

The optimization problems (38) and (42) are solved with the symbolic framework CasADi, and a primal–dual interior point solver IPOPT is selected to solve the nonlinear optimization problem [33].

B. Experimental Results: Nominal Safety

The RL control is tested together with a nominal safety filter in this experiment. The nominal model used in the filter is calibrated to fit the laboratory specifications and considers an average consumption, $d_{av} = 0.5$. The turnover computation is computed with the flow model (11b), with an average level of $h^* = 0.3[m]$ and a filter size $n_m = 24$. The whole control structure has multiple objectives, which, in some operational scenarios, can be conflicting objectives. This work has relaxed the constraints that represent the water age boundaries (38f) to prioritize safety at the tank level boundaries (38d) and (38e) during the transient.

During the first 5 h, the learning transient is observed in Fig. 8. The operation during this period is driven by the safety filter policy that often corrects the system trajectory, thus maintaining the safe operation of the system. During this first part, the identification of the Q-value function, and

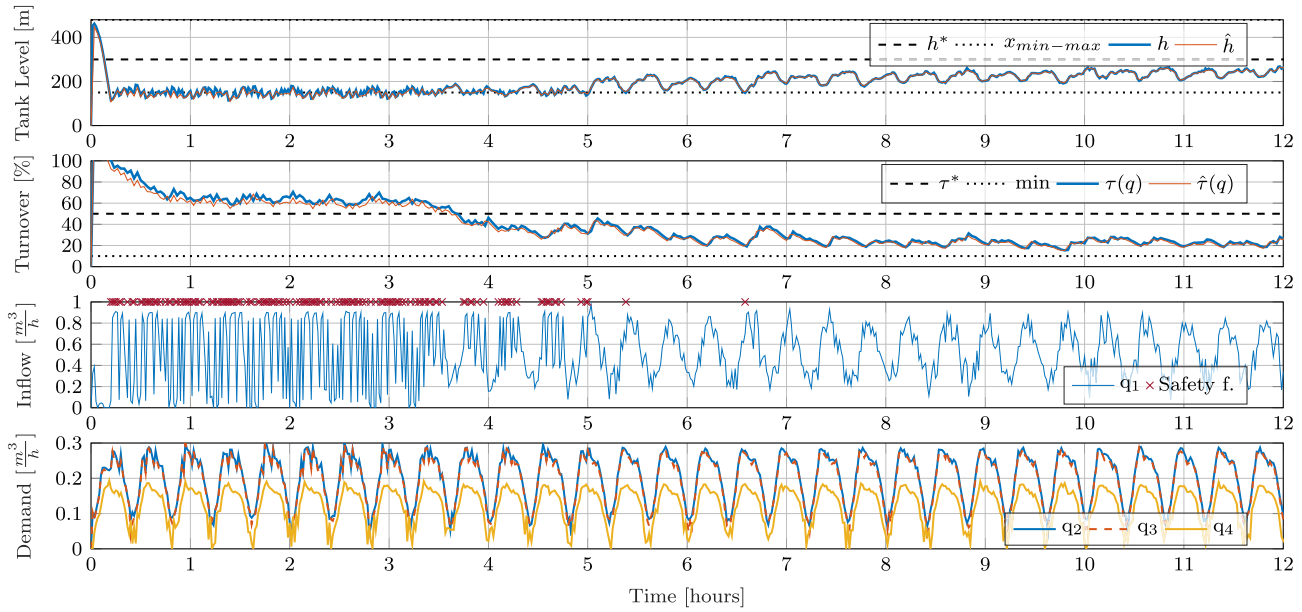


Fig. 8. Tank level regulation. Daily turnover of the tank, real signal τ . The inflow of the pumping station q_1 where the red crosses mark the actuations that are corrected by the safety filter. Demands from PZ1 (q_2 and q_3) and PZ2 (q_4), minutes in the test correspond to hours in reality.

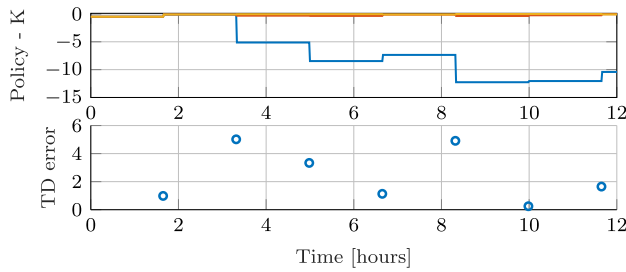


Fig. 9. Experimental results with RL control and nominal safety filter. Top: transient of the learned controller, $\hat{K}$. Bottom: TD.

subsequently, the optimal policy K , continues despite the system chattering around the safety boundaries; see Fig. 9. After 5 h of experiment, the RL controller improves, and the RL policy becomes dominant. Finally, the operation of the system is driven by a near optimal policy that slowly approached to a steady state.

C. Experimental Results: Combined Safety

In this experiment, the RL control is tested with a combined safety filter. In this case, the safety filter is challenged by assuming an inaccurate calibration of the nominal model, which considers 0.4 times the tank size A_{er} and 0.5 times the actual average demand d_{av} . The combined safety filter aims to compensate this poor model calibration. The GP model is initialized with a prior random input and output batches and the confidence gain $K_c = 1$.

The turnover model is computed with level measurements. The graph in Fig. 10 shows turnover signals computed with different models, and an error between the approximated flow and level models is observed. This error in the dynamics is especially patent in steady-state conditions, where $\hat{m}(q)$, that is computed with a constant d_{av} , fails to predict the periodicity of the disturbance. Fig. 11 shows the experimental results of an RL controller and a combined safety filter. The

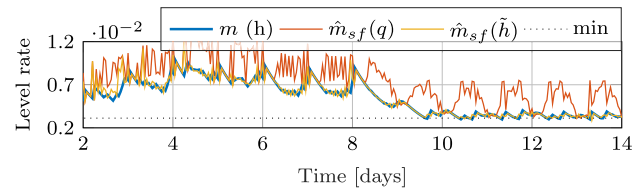


Fig. 10. Simulation of the positive level rate mean: blue signal is computed with model (13), and red and yellow are computed with the approximated model (14), with a nominal flow model (11b) and a predicted level (11a), respectively.

graph can be divided into two parts: a learning transient and a steady-state operation. The learning transient comprises a GP model learning and a Q-value function learning. During the first 4 h, when the GP model is not entirely trained, the predicted model differs from reality. See the bottom graph of Fig. 12. Therefore, the GP prediction provides a wide CI, which indicates high uncertainty in the prediction. This lack of confidence makes the safety filter provide a more conservative actuation. This effect is especially noticeable between hours 0 and 2, where the system path is corrected far from the safety boundaries. Once the GP model is improved, hours 3.5–6, the system can safely operate near the boundaries, hence enlarging the exploration limits. During the steady-state operation, after hour 6, the RL controller takes over system policy, and the safety filter is only triggered for correcting the approximated turnover signal $\hat{\tau}$. A scaling error is observed between the approximated and actual turnover signal $\hat{\tau}$ and τ . However, this error is considered a minor safety risk, since the approximated signal represents a worst case scenario from a model calibration.

The graph in Fig. 12 shows the convergence of Algorithm 1 to an optimal policy where the TD error is minimized. Likewise, the training of the GP model with Algorithm 2 shows a reduction of the prediction error between the nominal and combined models.

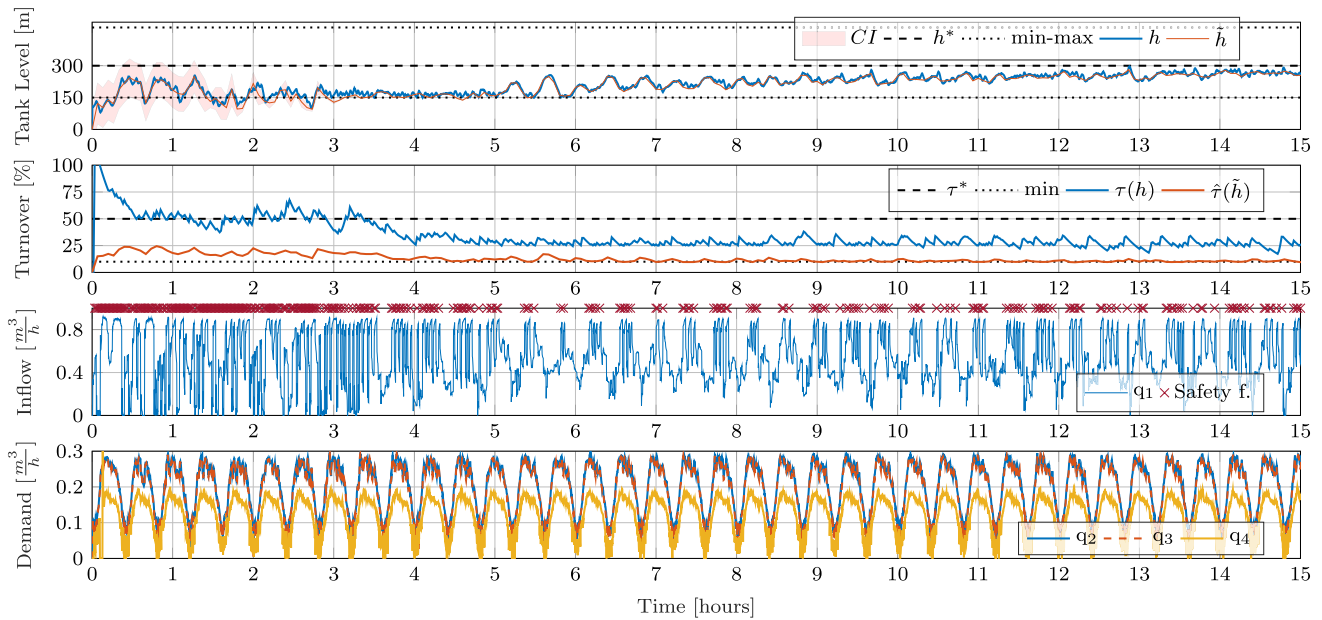


Fig. 11. Tank level regulation. Daily turnover of the tank, real signal τ and its approximation $\hat{\tau}$. Inflow of the pumping station q_1 where the red marks point the actuations that are corrected by the safety filter. Demands from PZ1 (q_2 and q_3) and PZ2 (q_4), minutes in the test correspond to hours in reality.

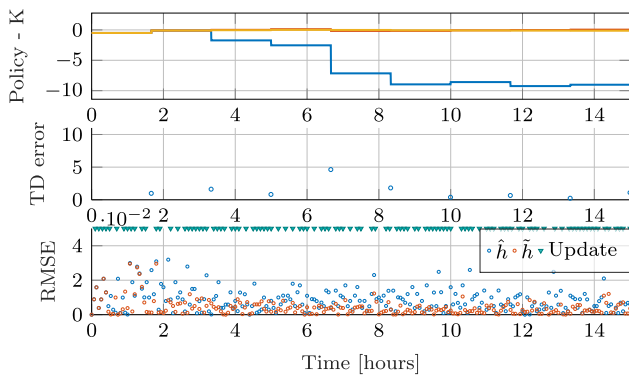


Fig. 12. Experimental results with RL control and combined safety filter. Top: transient of the learned controller, $\hat{K}$. Middle: TD computed as $|\hat{K}_t - \hat{K}_{t+1}|$. Bottom: root-mean-square error (RMSE) of the residual generated by the estimated $\hat{h}$ and $\hat{h}$ compared with the measured level h . The triangles show the GP model updates during the execution of Algorithm 2.

VI. DISCUSSION

Based on the collected results, a brief discussion of the strengths and weaknesses of the presented method is given.

A. Learning Controller

The presented RL method shows to be a suitable solution for obtaining an optimal-adaptive policy. The linear learning structure with polynomial bases provides a preliminary structure to the Q-value function that suits the linear behavior of the system and quadratic rewards to be learned. The design criteria in this controller prioritize the simplicity of the solution over performance, facilitating the adaptation to a changing environment. Having a simple learning structure increases the interpretability of the results compared with other black-box methods, thus reducing the commissioning time and cost. However, it restricts the management to simple objectives.

The performance and convergence of this linear controller rely on the assumptions, such that linear dynamics describe the system's behavior, quadratic rewards describe the management

objectives, and the operation is around a fixed operating point. This presents a challenge to validate the controller in real systems where the operation is uncertain, and the system nonlinearities, such as the actuator dynamics or communication delays, are more present in the control loop, as well as restricting the management objectives to simple tasks.

B. Safety Filter

The proposed RL optimization criteria do not specifically account for the physical limitations of the system. As a result, a safety filter is utilized to maintain safety in the operation. This filter overrules the RL policy to correct the system trajectory when approaching unsafe areas. Moreover, the use of a safety filter allows the inclusion of nonquadratic management objectives in the safety criteria, such as the water age. This control architecture suits well this industrial application compared with other solutions. For instance, having the safety encoded together with the learning controller increases the risks of failure, since the policy might explore unsafe areas before learning a safe policy. The experimental results, obtained under nominal and combined safety conditions, show the importance of including a safety filter for assisting a learning controller or any existing controller with uncertain policy.

Comparing the performance of both solutions is difficult, since the accuracy of prior system information is subjected to each case. In cases with accurate system information, the solution with GP regression might be unnecessarily complex. However, when dealing with uncertain environments, a combined safety solution shows an efficient robust performance.

The risk assessment of the RL policy is performed, including a variable CI . The confidence gain K_c modifies the learning transient on the application convenience. A high gain represents a lack of confidence in the RL policy and reduces the RL exploration area. However, poor state-space exploration might lead to suboptimal RL policies. On the other hand, a low gain relaxes the safety condition and allows the RL policies to

explore a wider state-space region. The feasibility of the safety control problem is an important factor to be considered when selecting a suitable K_c , and high gain increases the influence of the GP regression model, in particular, the variance model. When the GP model is not completely trained, this influence might be an issue, and the safety problem becomes unfeasible.

One of the purposes of having real-time training for the GP, Algorithm 2, is to decrease the CI gradually, thus removing the unnecessary safety assistance and entrusting the RL to drive the system's policy. In case of reaching large CIs that conflict with the safety optimization, the algorithm neglects the variance to find a safe-robust solution. The experimental results conclude that providing robust safety while learning the GP model is challenging. Insufficient training of a GP model brings a high uncertainty in the predicted mean and variance. This study reduces the variance dynamics (CI) impact by reducing the confidence gain K_c to facilitate the computation of a feasible solution. This safety method is a 1-step ahead optimization that does not imply significant computational effort. Moreover, the GP model utilizes the same collected data for learning the Q-value function x_k and u_k to facilitate the implementation.

Finally, a brief discussion analyzing the joint performance of the learning controller and safety block is given. RL and safety are decoupled blocks that have different control objectives. Therefore, the formulation of these objectives must be balanced, such that the filter is not dominant with respect to the existing controller and vice versa. The control actions' smoothness also conflicts with the learning performance that requires persistent variations to identify the Q-value function. Note that the RL method is based on a least squares estimate, which is sensitive to the quality of the collected data, and a persistence of excitation signal must be present also during safety scenarios.

VII. CONCLUSION AND FUTURE WORK

This article proposes an optimal-adaptive control solution for a WDN with an elevated reservoir. The management of this network comprises two main operation objectives: controlling the network pressure and water age by regulating the tank's level and turnover. The control method consists of an RL linear controller that provides the primary policy and a policy supervisor that predicts and corrects potential violations of the safety boundaries. The predictions are based on a linear nominal model. The performance of this filter is subjected to the calibration of the model. Therefore, a GP regression is included in the filter to compensate for incorrect model calibration and include nonlinearities that a linear model cannot capture. The proposed control strategy is validated in a laboratory setup that emulates the WDN dynamics. The experimental results reflect the benefits and the limitations of the method.

Although this article presents promising results that support the implementation of safe learning techniques in industrial applications, this method is challenged by several factors. Future development of this method must consider a nonlinear learning architecture that allows the formulation of nonlinear objectives in the reward. It allows learning an optimal policy

that includes important management objectives, such as water age or operational costs. Furthermore, there is potential to streamline the commissioning of the safety filter in different systems by reducing the reliance on heuristically adjusted parameters.

APPENDIX GP REGRESSION

This model is inspired by the formulation presented in [34] where the uncertainty between an imperfect nominal model and the real system is modeled with a GP regression. Then, by applying a GP regression, this method aims to learn the uncertainty term $r(z)$ represented in (39) as follows. First, the residual is expressed as follows:

$$y_k = r(x_k, u_k) + w_k = B_r^\dagger \left(\underbrace{h_{k+1}}_{\text{measure}} - \underbrace{\hat{f}(x_k, u_k)}_{\text{nominal}} \right) \quad (45)$$

where $B_r^\dagger$ is the Moore–Penrose pseudo-inverse of B_r . Then, consider a training dataset $\mathcal{D}$ that consists of M observations

$$\mathcal{D} = \{\mathbf{y} = [y_1, \dots, y_M]^T \in \mathbb{R}^M, \mathbf{z} = [z_1, \dots, z_M]^T \in \mathbb{R}^{M \times n_z}\} \quad (46)$$

where z denotes an input vector and y a scalar output (target). This definition is performed by assuming that each of elements y of the output vector is independent for the given input data z_k . Then, by giving a GP prior on r with kernel $k(\cdot, \cdot)$, prior mean function is zero

$$y \sim \mathcal{N}(0, K_{zz} + I\sigma^2). \quad (47)$$

The result is a normally distributed measurement where K_{zz} is the Gram matrix of the data points, such that $K_{ij} = k(z_i, z_j)$, and the selection of the kernel k structure and its parameterization determines the distribution of the predicted output. In this article, the selected kernel is the squared exponential kernel function

$$k(z_i, z_j) = \sigma_f^2 \exp\left(-\frac{1}{2}(z_i - z_j)^T L^{-1}(z_i - z_j)\right) \quad (48)$$

where L is a positive diagonal length scale matrix and σ_f^2 the signal variance. This kernel is selected based on the domain knowledge, since the dynamics of the system present a continuous and smooth behavior.

The joint distribution of the training data $\mathbf{z}$ and the test data z_* is

$$\begin{bmatrix} \mathbf{y} \\ y_* \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} K_{zz} + I\sigma^2 & K_{zz_*} \\ K_{z_*z} & K_{z_*z_*} \end{bmatrix}\right) \quad (49)$$

where $[K_{z_*z}]_j = k(\mathbf{z}_j, z_*)$, $K_{z_*z} = K_{z_*z}^T$, and similarly $K_{z_*z_*} = k(z_*, z_*)$. The resulting conditional distribution of the uncertainty residual is Gaussian [35]

$$\hat{r}(y_*|\mathbf{y}) = \mathcal{N}(\mu^r(z_*), \Sigma^r(z_*)) \quad (50a)$$

$$\mu^r(z_*) = K_{z_*z}(K_{zz} + I\sigma^2)^{-1}\mathbf{y} \quad (50b)$$

$$\Sigma^r(z_*) = K_{z_*z_*} - K_{z_*z}(K_{zz} + I\sigma^2)^{-1}K_{z_*z} \quad (50c)$$

where $\mu^p(z_*)$ and $\Sigma^p(z_*)$ are mean and variances of the GP.

Algorithm 2 Training of the GP Model

```

1: Input:  $n_{gp}$ ,  $e^*$ 
2: Initialisation:  $[\sigma_{f0}, L_0, \sigma_0] \leftarrow \text{fitrgp}(\mathbf{z}_0, \mathbf{y}_0)$ 
3: repeat at every iteration  $k = 1, 2, \dots$ 
4:   collect data  $\hat{h}_k, z_k$  and  $y_k$ 
5:    $e_k = \text{RMSE}(h_k - \hat{h}_k)$ 
6:   if  $e_k \geq e^*$  then ▷ Collect data
7:     save  $z_k$  and  $y_k$  in stack  $\mathbf{z}_j$  and  $\mathbf{y}_j$ 
8:     if  $k = (j+1)n_{gp}$  then ▷ GP update
9:        $[\sigma_f, L, \sigma] \leftarrow \text{fitrgp}(\mathbf{z}, \mathbf{y})$ 
10:       $j = j+1$ 
11:   end if
12: end if
13: until

```

The GP model is trained with measured data that are collected by executing Algorithm 2 in real time. These data batches are built with the vector structure (31), which combines the augmented state vector x and control action u . The control scheme used for both training and operation is shown in Fig. 3, and the policy is initially arbitrary and gradually stabilizes. The algorithm needs to be initialized with prior data $\mathbf{z}_0$ and $\mathbf{y}_0$, the number of new samples per update n_{gp} , and the threshold e^* . This threshold represents the minimum deviation between the measured variable h_k and estimated $\hat{h}_k(z_k)$, with $z_k = [x_k^T, u_k^T]^T$. The hyperparameters of the GP are obtained with the MATLAB function `fitrgp()`. The collected data are stored between updates in a last in first out stack (LIFO).

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